Attractor Representations for Manifolds in Neural Circuits

For Application to the SWC-GCNU PhD Programme

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Agenda

Motivation

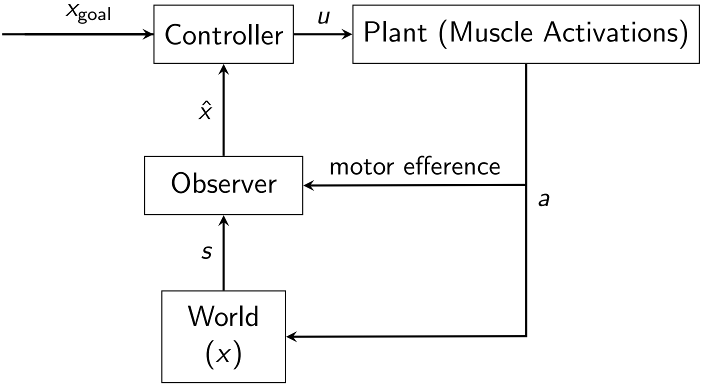
The brain is thought to hold internal representations of variables, in order to solve the below problem:

$$\text{Given } x_t, C, \text{ find } a_t^* \sim f(x_t, C) \quad \text{such that } \sum_{\tau=t} R(x_\tau, a_\tau^*, C) \text{ is maximised.}$$

where x_t is a current state, C is the context (which may contain x_{goal}), and a_t^* is the action. R is the reward function, which can be additionally dependent on $\{\gamma_\tau\}_\tau$.

Motivation

In a classical control perspective, we only observe low quality s from the world and hence must have an internal state estimate.



Motivation

Now, we must notice that x_t can live in different spaces!

- Spatial cognition: 2D space, *grid cells, place cells*
- Head orientation: 1D ring, *HD system in mice and fly*
- Head *rotation*: SO(3), *bat presubiculum, mice SC (?)*
- Object manipulation: CGA, *motor hierarchy (?)*

... and various graph structures related by discrete actions (see Whittington et al. 2022 Nature).

When the state space is continuous and locally Euclidean, they are *manifolds* and we denote $x_t \in \mathcal{M}$.

Agenda

CAN Problem Setup

We use a set of neural states $g_t \subset \mathbb{R}^d$ to *represent* x_t. Described in general terms,

$$\dot{g} = F(g, u, t) \tag{1}$$

where $u \in \mathbb{R}^{d_{\mathcal{M}}}$ lives in the tangent space of the manifold.

This continuous time system is often approximated by the following discrete time system, and we assume that the system is invariant:

$$g_{t+1} = \mathcal{F}(g_t, u_t) \tag{2}$$

Ring Attractor Theory (Zhang, 1996)

Similar to the classical setup, investigate the following system (e.g. Amari, 1972)

$$g = \sigma(v)$$
$$\tau \dot{v} = -v + W(t) * g$$

where $v, g \in \mathbb{R}^d$ are the neural activity (e.g. voltage) and the neural state after nonlinearity. $\sigma : \mathbb{R}^d \to \mathbb{R}^d$

$W(t) \in \mathbb{R}^{d \times d}$ gives the weight distribution to be convolved with the neural state.

How was the ring attractor set up?

Parametrize neural response as $g(\theta, t)$ [1]

Induce rotation invariance by a circulant weight kernel $W(\theta, \theta', t)$ that only depends on the difference $\theta - \theta'$.

$$W * g(\theta, t) \to \int_{\Theta} W(\theta - \theta', t) g(\theta', t) d\theta'$$

Then the weight distribution is a function defined in the angular domain, $W(\theta, t) = W_{\text{odd}}(\theta, t) + W_{\text{even}}(\theta, t)$

When $W_{odd} = 0$, the system can settle to some static attractor state with appropriate σ (minimum energy in the Lyapunov form).

[1] Consider infinite neural coding resolution

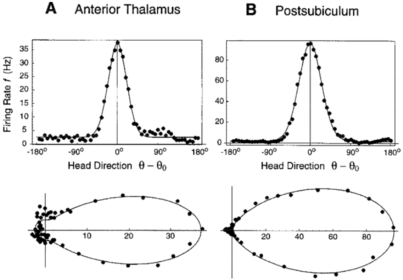
Attractor network as a solution

Candidate directional tuning curve: [2]

$$g(\theta) = A + B\exp(K\cos(\theta - \theta_0))$$

Candidate nonlinearity: [3]

$$\sigma(v) = a\ln^{\beta}(1 + \exp(b(v + c)))$$



[2] Circular normal distribution, parameters fitted from observed neural firing

[3] Exponential decay behavior in the negative half and power function in the positive half

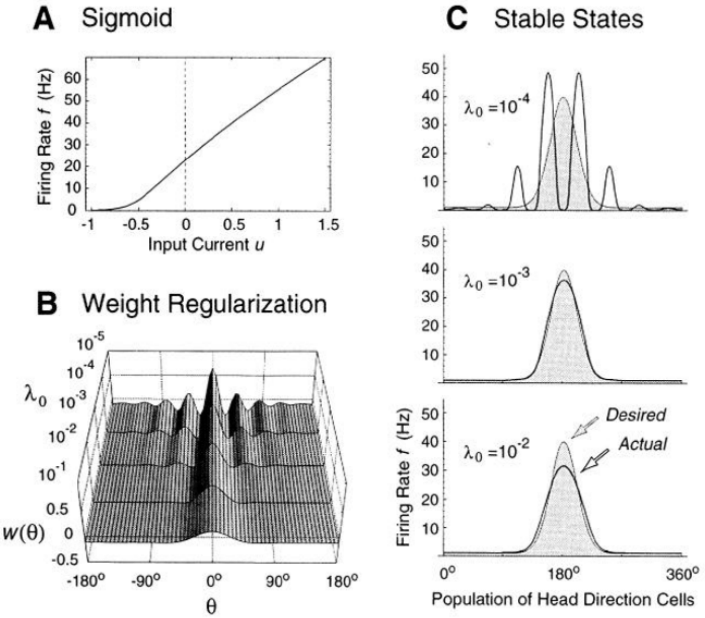
Solve for weight distribution in the Fourier domain:

$$\dot{v}(\theta) = -v(\theta) + \int_{\Theta} W_{\text{even}}(\theta - \theta')g(\theta', t)d\theta' = 0$$
$$v(\theta) = W * g(\theta)$$
$$\hat{v}_n = \hat{W}_n \hat{g}_n \quad \forall n \in (-\infty, \infty)$$

Adding a small weight regularization to encourage finite-energy solution ($\|W\|_2^2 < \infty$), we can solve for the Fourier coefficients of the weight distribution.

$$\hat{w} = \arg\min_{\hat{w}} \sum_n |\hat{v}_n - \hat{w}_n \hat{g}_n|_2^2 + \lambda |\hat{w}_n|_2^2$$
$$\hat{w}_n = \frac{\hat{v}_n \hat{g}_n}{|\hat{g}_n|_2^2 + \lambda}$$

Optimized Weight Distribution



Shifting mechanism

To achieve rigid bump shape when shifting along the manifold, we must ensure that the odd component of the weight distribution is its derivative [4].

$$W(\theta, t) = W_{\text{even}}(\theta) + \gamma(t)\frac{dW_{\text{even}}}{d\theta}$$

The relationship between $\gamma(t)$ and angular velocity $\omega(t)$ is given by the following equation [5]:

$$\gamma(t)/\tau = \omega(t)$$

[4] This can be shown by dissecting a symmetric travelling profile

[5] This is to relate the speed at which the rigid profile is rotated to the

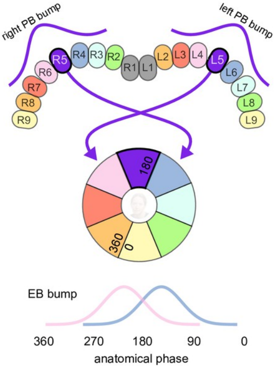
Biological implementation of the multiplicative gate

$W(\theta, t)$ describes the total weight distribution, which can be realized by two groups of cells:

Rep : Component of the cell population with symmetric connections, implementing $W_{\text{even}}(\theta)$

Input : Component of the cell population with asymmetric connections, effectively implementing $\frac{dW_{\text{even}}}{d\theta}$, dual tuned to the velocity signal.

This is called the copy-and-shift mechanism, and is almost directly implementable in the fly brain.



[a]

[a] Hulse et al., 2021

Attractor optimized for cells in the EC

In the above formulation, cells are defined by tuned angle.

When considering other systems (e.g., mammalian), it becomes less and less easy to do this type of definition.

EC = {grid cells, border cells, band-like cells, etc.}

Cueva and Wei (2018): Optimize directly for path-integration.

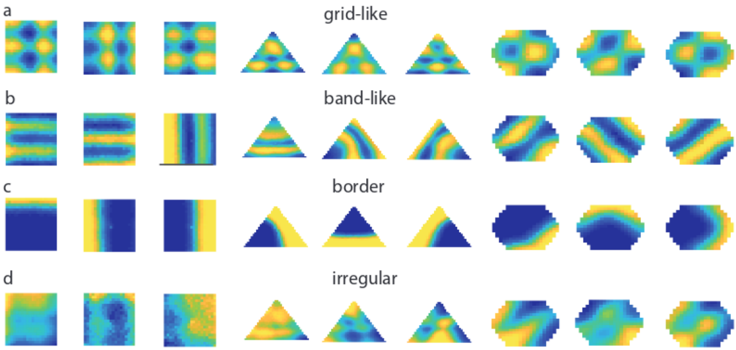
$$\tau \frac{dv_i(t)}{dt} = -v_i(t) + \sum_j W_{ij} g_j(t) + \sum_k^{N_{\text{input}}} W_{ik}^{\text{input}} u_k(t) + b_i + \epsilon_i(t)$$

$$y_j(t) = \sum_i^N W_{ji}^{\text{output}} g_i(t)$$

for loss function

$$\mathcal{L} = \langle \| y - y_{\text{target}} \|^2 \rangle_{T, N_{\text{out}} B} + \lambda_1 \langle \| W \|_2^2 \rangle + \lambda_2 \langle \| g \|_2^2 \rangle$$

Training Results



Agenda

CAN Problem Setup

For a neural dynamical system to be a CAN for a manifold, it should satisfy the following two postulates.

$$\text{Stability:} \quad \mathcal{F}(g_t, 0) = g_t \tag{3}$$

$$\text{Path Integration:} \quad \int_t^{T_1} u_\tau \, d\tau = \int_t^{T_2} \mu_\tau \, d\tau \iff g_{T1} \mid \{u_\tau\}_{\tau \in [t, T1]}, g_t = g_{T2} \mid \{\mu_\tau\}_{\tau \in [t, T2]}, g_t \tag{4}$$

where integration is performed on the manifold, such as by consulting its Lie algebra.

Data and Training

Trajectories of actions are sampled by an AR(1) process

$$u_{t+1} = \alpha u_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \Sigma_{\mathrm{AR}})$$

With this base dataset of trajectories, each gradient step is expressed as following.

$$\Phi_{e+1} = \mathcal{D}(\Phi_e, \mathbf{A}_e, \mathbf{P}_e)$$

where $\mathbf{A} = \{u_{1:T}^{(i)}\}_{i \in 1:B}$ and $\mathbf{P} = \{\mu_{1:T}^{(i)}\}_{i \in 1:B}$, such that

$$\| \int_1^T u_\tau^{(i)} d\tau - \int_1^T \mu_\tau^{(i)} d\tau \| < \sigma_{\mathrm{x}} \quad \forall i$$

Loss Function

Similarity:

$$\left\| \int_t^T u_\tau \, d\tau - \int_t^T v_\tau \, d\tau \right\| < \sigma_{\mathrm{x}} \implies \mathcal{F}(g_t, u_{t:t+T}) \approx \mathcal{F}(g_t, v_{t:t+T})$$

Contrast:

$$\int_t^T u_\tau \, d\tau \neq \int_t^T v_\tau \, d\tau \implies \mathcal{F}(g_t, u_{t:t+T}) \neq \mathcal{F}(g_t, v_{t:t+T})$$

Total loss:

$$\mathcal{L}_{\text{contrastive}} = \frac{\lambda}{\beta} \sum_{i=1}^{B} \phi(g_{t^\star}^{\mathbf{A},(i)}, g_{t^\star}^{\mathbf{P},(i)}) - \log \sum_{k=1}^{B} \exp(\frac{\phi(g_{t^\star}^{\mathbf{A},(i)}, g_{t^\star}^{\mathbf{P},(k)})}{\beta})$$

Architecture

Segment the neural net into Rep and Input components, as in the Zhang model.

$$g_{t+1} = \begin{bmatrix} g_{t+1}^{\text{rep}} & g_{t+1}^{\text{in},1} & \dots & g_{t+1}^{\text{in},\mathrm{J}} \end{bmatrix}^{\top}, \quad \text{where} \tag{5}$$

$$g_{t+1}^{\text{rep}} = \sigma(W_{rr} g_t^{\text{rep}} + \sum_{j \in [1,\mathrm{J}]} u_t^j W_{ir}^j g_t^{\text{in},j} + b_r)$$

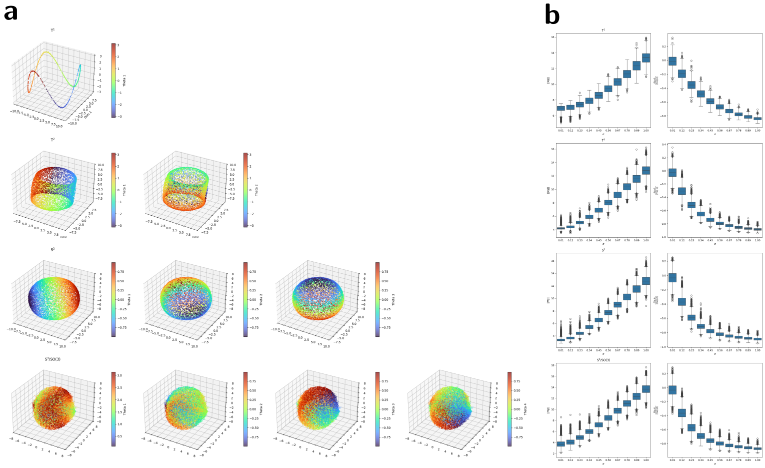
$$g_{t+1}^{\text{in},j} = \sigma(W_{ri}^j g_t^{\text{rep}} + \sum_{j \in [1,\mathrm{J}]} u_t^j W_{ii}^j g_t^{\text{in},j} + b_i^j)$$

A more flexible setup can be used for a similar path integration problem, allowing for more complex modulation by actions.

$$W(u_t) \triangleq \mathrm{MLP}(u_t)$$

Results

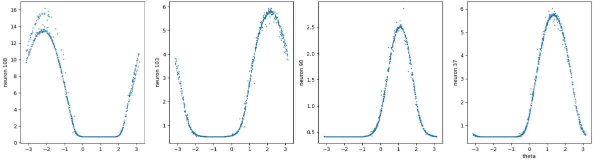
Check structure and noise robustness of the learned attractor.

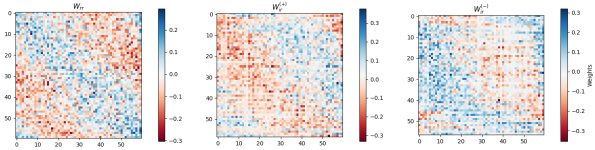


Comparison with Zhang's model

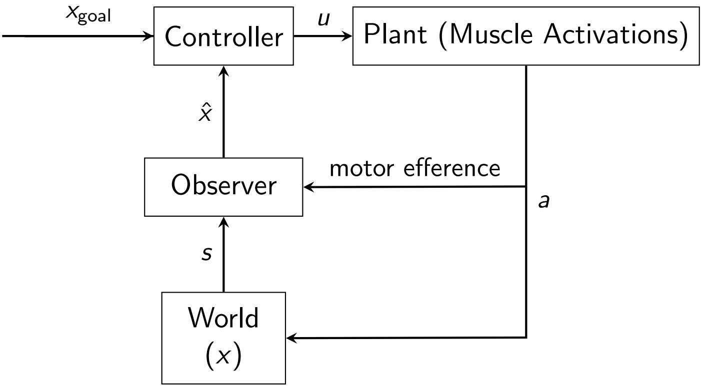
Additional constraints:

- Weight regularization
- Biologically plausible implementation (an input channel for each sign)

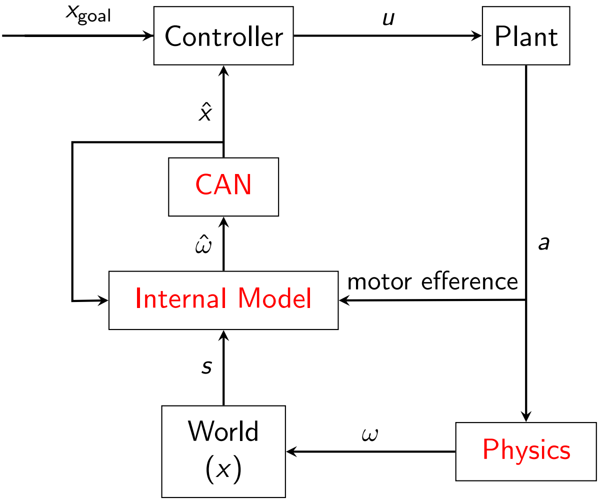




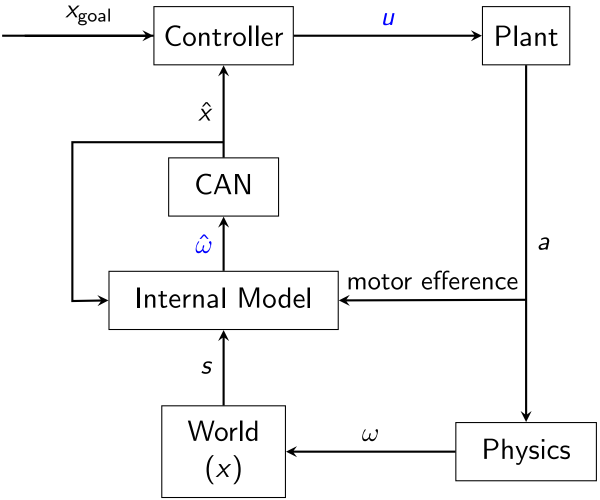
Revisit the Controller



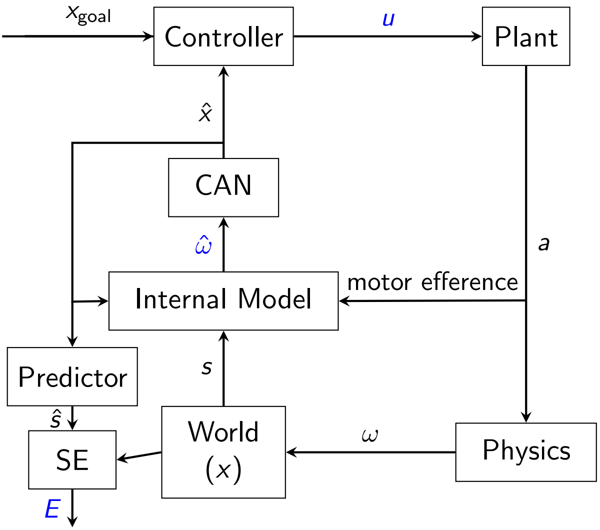
Using the attractor for motor control



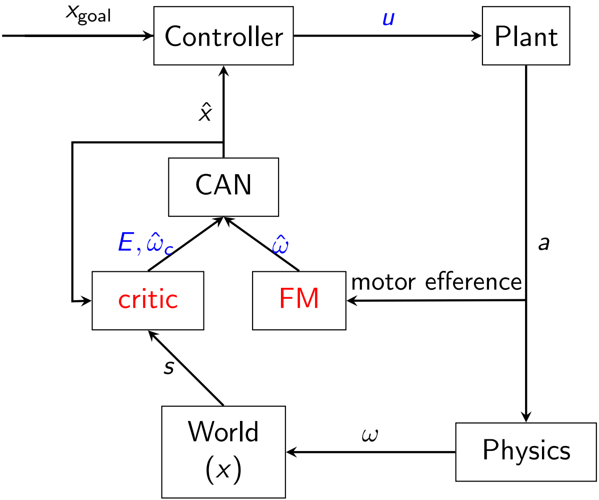
Training the attractor for motor control



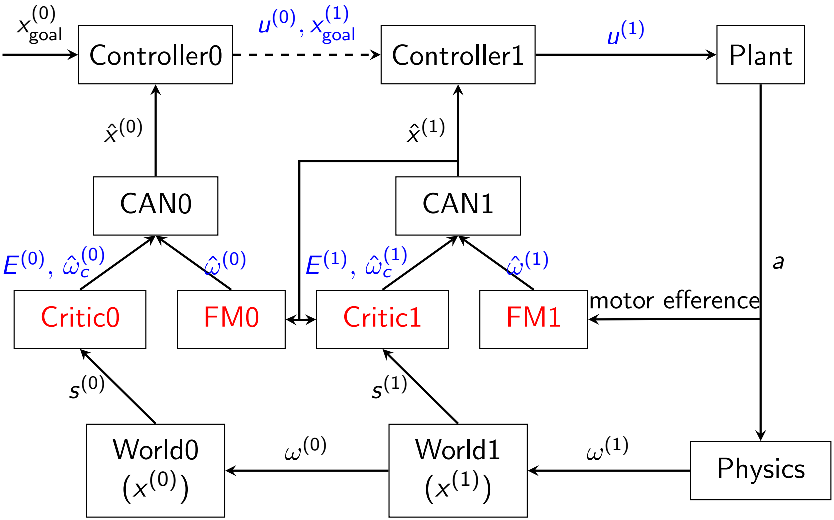
Training the attractor for motor control 2



Training the attractor for motor control 3



Extension: Hierarchical Attractors for Motor Control



Summary

- Attractor networks are a powerful tool for representing manifolds in neural circuits.
- They can be trained with minimal assumptions, and can be used for motor control.
- The attractor network may be used as an observer for the stimulus manifold in motor control. The full system may be mapped onto a neural circuit for a control problem.
- My next steps, after training the control circuit, include experimenting with biologically plausible implementations of learning rules. In a longer term, one may consider chained attractor networks for hierarchical control.

Proof of the shift mechanism

let $v(\theta, t) = V(\theta - \omega t)$ be the symmetric travelling profile.

$$V(\theta) + \gamma \frac{dV}{d\theta}(\theta) = [W_{\text{even}} + W_{\text{odd}}] * g(\theta)$$

$$V(\theta) = W_{\text{even}} * g(\theta)$$

$$\gamma \frac{dV}{d\theta}(\theta) = W_{\text{odd}} * g(\theta)$$

$$W * f = \gamma \frac{dW_{\text{even}}}{d\theta} * f$$